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TECHNOLOGICAL
UNIVERSITY**
SINGAPORE

HE4045 Quantitative Economics with Data Science Final Project

*Research Question: How Does Property Tax Revenue Relate to
Housing Affordability, Economic Growth, and Standard of Living?*

Group 11

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1. Introduction

Singapore's property market is unique, famous for public housing that houses 77.4% of residents. However, despite Singapore having one of the highest home ownership in the world (*MND | Public Housing*, n.d.), there is increasing attention on the soaring property prices, given that Housing Development Board (HDB) resale flats rose 9.6% in 2024, which is almost twice the year before, while the private market also saw a rise of 3.9% (Foo, 2025). We have been seeing more efforts by the government to cool the housing market.

Cooling measures, such as reducing the maximum amount of loans that buyers can get for their HDB flats, aim to prevent overheating of the public housing market. By increasing the duty for foreign property buyers, they intend to relax the private housing market. These efforts signal the government's intent to cool off the property market, and also shows the restrictions in the property market.

The property market is significant, since it reflects one of the fundamental needs of humans, which is shelter. While property, being a significant asset to most Singaporeans, continues to closely relate itself to the standard of living of locals. Thus, the property market would reveal strong information about the economy and people's lives.

All these measures have added to speculations of a housing affordability crisis in Singapore. Thus, we are seeing a rising importance in analysing how local policies affect markets, and by extension, the economy. Property tax is a significant part of the government's income and has widespread influence on the housing market affordability and even economic activity. Property tax can also further influence urban development as people and firms move and conduct activities in different areas.

In this project, we will be looking at private property housing and the relation between property tax revenue, housing affordability and standard of living in Singapore.

2. Scope and Objective

Property tax revenue represents part of the earnings of the government. It reflects government sentiment and is a policy that can affect the property market and the economy. Thus we use it to examine government earnings and expenditure.

Private property is significantly more expensive in Singapore as compared to public housing. Public housing prices in Singapore remain highly regulated by the government, thus the public housing market faces many restrictions in order to keep public housing accessible to most Singaporeans. Thus, we will only focus on private housing which includes landed property and condominiums, since it resembles a free market more due to it being less controlled by the government and thus a more accurate reflection of the effect of changes in property tax revenue.

As for time range, subject to data availability, we will narrow the scope to 2010 to 2023, which allows us to focus our study in a manageable and recent scope. This allows our study to be relevant and consistent.

In this paper, we will first evaluate the relationship between levels of property tax revenue and housing affordability in Singapore, by examining the correlation between property tax revenue and property price-to-income ratio. We will use the price-to-income ratio as a proxy for housing affordability, which measures the median home price of the type of housing to median household income in Singapore. This would shed more light on whether the speculations about a housing affordability crisis is valid, and how the affordability of housing changed in the time period used.

We will then analyse the relationship of the property tax revenue and standard of living in Singapore. To proxy the standard of living, we will use Gross Domestic Product (GDP) per capita and Human Development Index (HDI). This aims to reveal how property tax rates affect the people's everyday lives and if the positive effects of tax rates, such as increased government provisions on public facilities, outweigh the negative effects of tax rates for the people due to the loss in their income. While also revealing how standard of living affects property tax revenue.

3. Methodology and Approach

This study adopts a quantitative approach to examine how property tax rates affect private housing affordability and the overall standard of living in Singapore. The study period spans from 2010 to 2023, in order to preserve the validity and relevance of the observation, on top of data availability factors.

Data for the analysis includes:

- Property tax revenues, separated into landed property and condominium tax categories (Annual Value and Property Tax by Property Type | IRAS, n.d.).
- Housing affordability, computed as the ratio of average housing prices to median income. As only a housing price index (Private Residential Property Price Index (Base Quarter 2009-Q1 = 100), Quarterly | URA, n.d.) was available, average prices were estimated using base-year values adjusted by the annual price index.
- GDP per capita, used to represent income levels and standard of living (World Bank Open Data, n.d.).
- Human Development Index (HDI), to capture broader social and quality-of-life dimensions beyond income (United Nations, n.d.).

The analysis consisted of three main stages. First, descriptive statistics and trend analyses were conducted to observe the evolution of property tax rates and housing affordability over time. Second, affordability indices were calculated to identify how changes in property tax levels correspond to shifts in housing affordability. Third, correlation analysis was performed to measure the strength and direction of relationships among property tax, GDP per capita, and HDI.

The study hypothesizes that higher property taxes are associated with lower housing affordability, as they increase ownership costs, but potentially higher standards of living, as they contribute to government revenue and public welfare.

4. Findings and Analysis

4.1 Trend of Landed House Affordability



Figure 1: Average Price of Landed House to Median Income Over Time

Figure 1 illustrates the trend of the price-to-income ratio from 2010 to 2023, representing private housing affordability in Singapore. The ratio increased from 4.9 in 2010 to a peak of around 5.2 in 2011, indicating declining affordability as property prices rose faster than income. This was followed by a gradual decline between 2012 and 2017, where the ratio dropped to approximately 3.8, suggesting improved affordability during this period. However, from 2018 onwards, the ratio began to rise again, reaching close to 4.8 by 2023.

Between 2012 and 2017, Singapore’s landed housing affordability improved markedly as reflected by a declining price-to-income ratio. This period coincided with the introduction of strong government cooling measures such as the Total Debt Servicing Ratio (TDSR) framework (Phang, 2025) and higher Additional Buyer’s Stamp Duty (ABSD). These policies restricted speculative and leveraged property purchases by capping the proportion of income that could

be used for loan repayment and raising the transaction cost for second-home and foreign buyers. As a result, the pool of potential buyers for landed properties shrank, reducing speculative demand and tempering price growth. At the same time, median household incomes continued to rise steadily, further improving the affordability ratio. Economically, this trend illustrates how binding credit and liquidity constraints effectively shifted the demand curve for high-value properties leftward, moderating prices even without large changes in supply.

However, from 2020 onwards, affordability declined again, with the ratio rising sharply due to a combination of pandemic-driven demand shocks and persistent supply limitations. The COVID-19 pandemic reshaped household preferences toward larger, low-density homes, leading many high-income and asset-rich individuals to seek landed properties for greater space and privacy. This sudden surge in demand collided with a highly inelastic supply base, as construction delays and Singapore's limited land availability restricted new landed housing stock. Simultaneously, global liquidity conditions and low interest rates increased asset valuations, allowing wealthy buyers to absorb higher prices with relative ease. Consequently, even under continued cooling measures, landed property prices rose faster than incomes, worsening affordability.

4.2 Trend of Property Tax

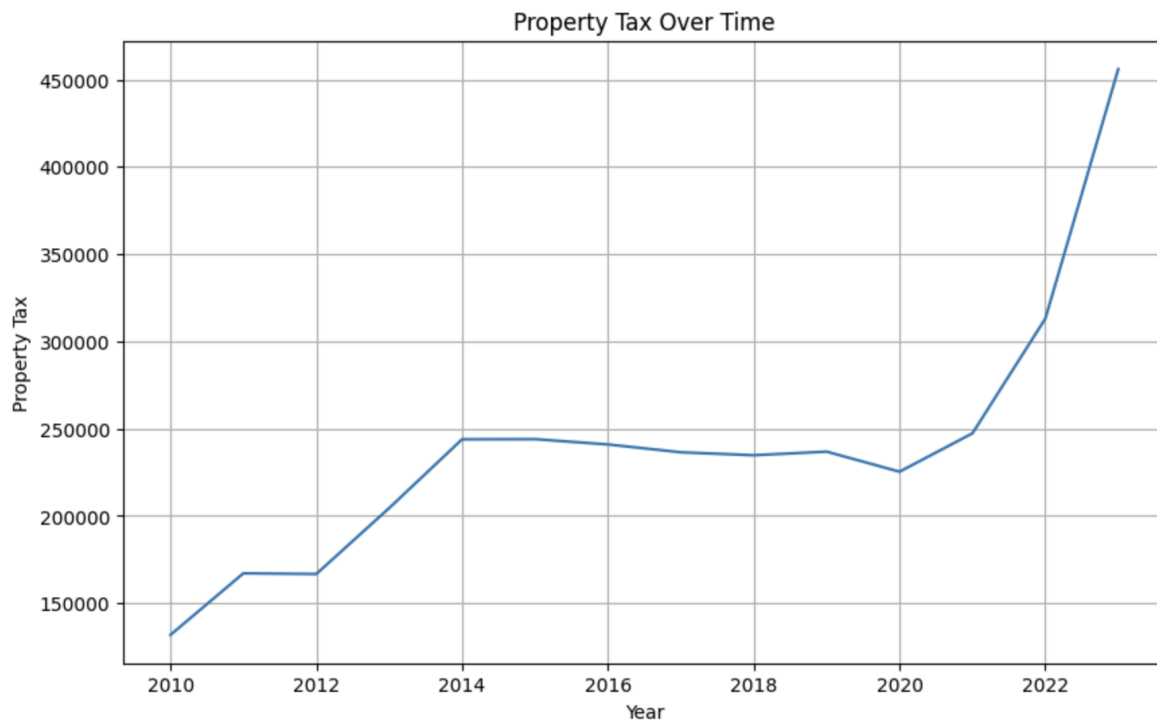


Figure 2: Property Tax Revenue from Landed House Over Time

Figure 2 shows the trend of property tax revenue from 2010 to 2023. Property tax collection increased significantly from around 130,000 in 2010 to over 450,000 by 2023. The steady growth until 2014 was followed by a period of relative stagnation between 2015 and 2019. After 2020, property tax revenues accelerated sharply.

From 2010 to 2014, property tax revenues for landed homes in Singapore rose gradually in tandem with increasing property values. This reflected growth in Annual Values (AVs), the assessed rental value used as the tax base. Rising market rents and sustained capital appreciation in the early 2010s pushed AVs higher, leading to greater tax collections even before any rate revisions. During this phase, buoyant housing demand and rapid appreciation in landed prices meant that higher property taxes were primarily a result of valuation updates rather than policy tightening.

Between 2015 and 2019, property tax levels stabilized as the impact of earlier cooling measures took hold. With speculative demand reduced and transaction activity slowing, landed property prices and rents plateaued, resulting in relatively stable AV assessments.

The period after 2020, however, marked a sharp upturn in property taxes for landed homes. As the pandemic spurred unprecedented demand for larger residences, market valuations and AVs surged once again. Concurrently, the Singapore government introduced progressive property tax reforms (Phang, 2023), raising marginal rates for higher-value properties to enhance tax equity. These dual forces, valuation increases and higher marginal tax rates, amplified total tax liabilities, especially for luxury and investment-grade landed properties. Economically, this reflects both a valuation effect (rising AV base) and a policy-driven incidence effect (greater progressivity). While higher property taxes increase the user cost of holding landed properties, the short-run price impact remains limited because of high supply inelasticity and concentrated ownership among high-income households. Thus, even as tax burdens rose sharply, affordability continued to decline, showing that recurrent taxation functions more as a redistributive and fiscal tool than as an effective short-term lever for price stabilization in the landed housing segment.

4.3 Relationship Between Property Tax and Affordability for Landed House

Pearson correlation coefficient between Price-to-Income Ratio and Property Tax: -0.0844881691374001

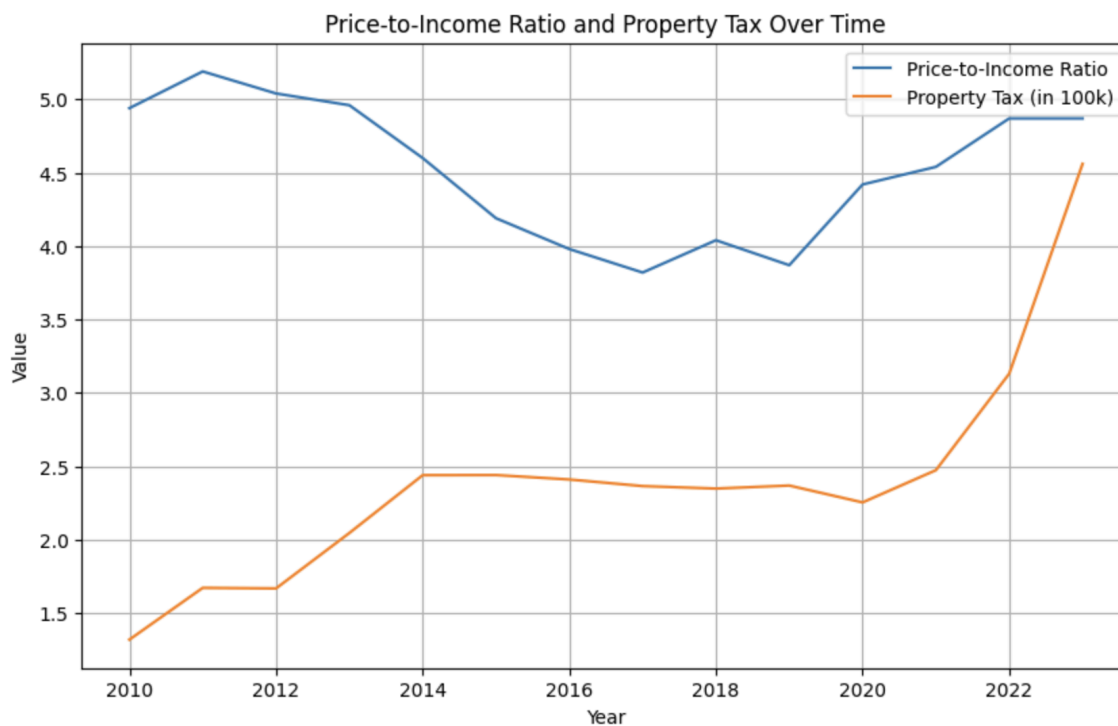


Figure 3: Correlation Between Property Tax and Affordability for Landed House

The correlation between property tax and housing affordability in Singapore's landed housing segment appears weakly negative, as reflected by the Pearson correlation coefficient of -0.084 . While this suggests that higher property taxes may be associated with slightly improved affordability, the relationship is far too weak to indicate any strong or direct effect. Economically, this outcome can be explained by the dominant role of non-price determinants in the landed housing market, such as wealth accumulation, foreign capital flows, and fixed land supply, relative to the incremental influence of recurrent property taxation.

In theory, higher property taxes should reduce demand by raising the user cost of ownership, discouraging speculative purchases and cooling price growth. This mechanism typically works well in markets with more elastic supply or broader buyer bases. However, Singapore's landed property market is exceptionally constrained and characterized by supply rigidity and high-income buyers. The total stock of landed homes is legally capped and grows very slowly due to land scarcity and zoning restrictions. As a result, even substantial increases in tax rates

or Annual Values exert limited influence on price formation, because supply cannot adjust and demand from wealthier households remains relatively insensitive to marginal cost increases.

4.4 Relationship Between Property Tax and Affordability for Condominium

... Pearson correlation coefficient between Price-to-Income Ratio and Property Tax for Condo: -0.19082795725829915

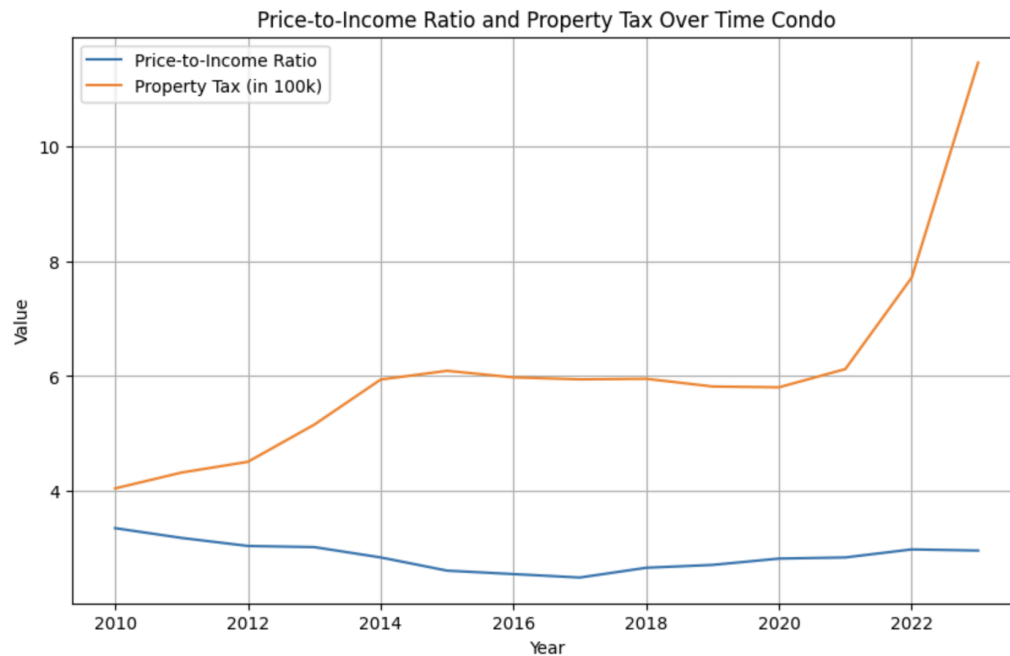


Figure 4: Correlation Between Property Tax and Affordability for Condominium

Compared with landed housing, condominiums show a more elastic response to policy and tax changes. The landed segment remains dominated by wealthier buyers and constrained supply, making it less responsive to taxation. In contrast, the condominium market, larger and more accessible, responds more visibly to regulatory measures, meaning that while property taxes alone have limited impact, they are relatively more effective in stabilizing affordability within the condo market than in the landed housing segment.

4.5 Relationship Between Property Tax and Standard of Living

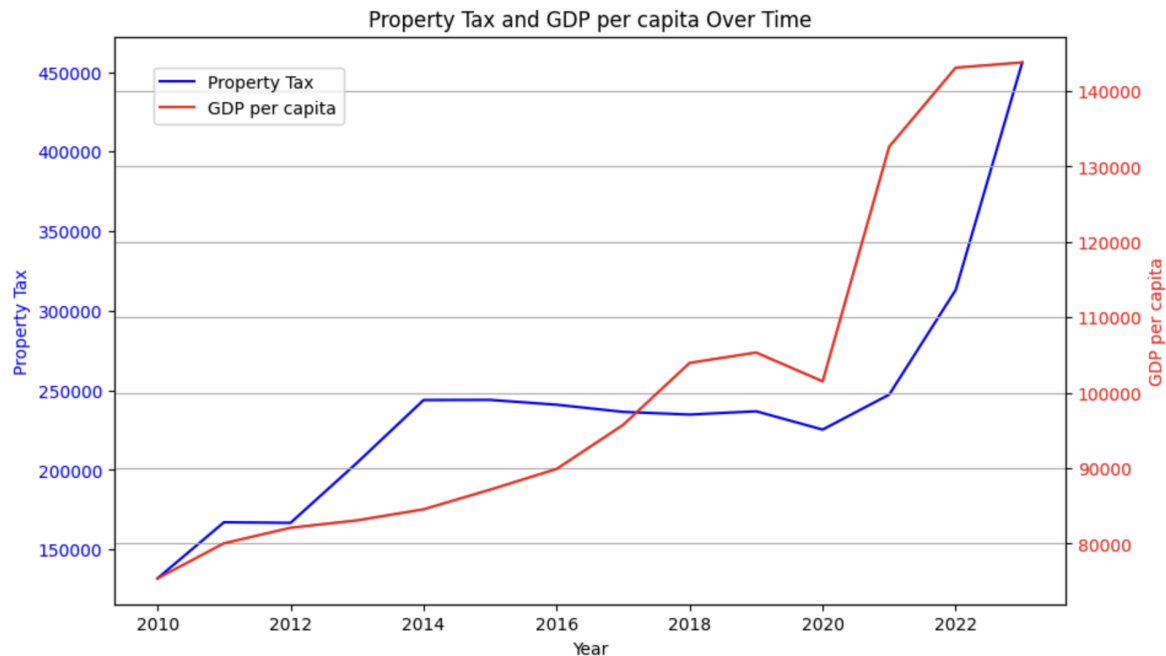


Figure 5: Correlation Between Property Tax and GDP per Capita Over Time

Figure 5 reveals an interesting positive co-movement between property tax revenue and GDP per capita from 2010 to 2023. The upward trajectory of both variables suggests that as property tax collections rise, Singapore's economy and average income levels also strengthen. Economically, this can be interpreted through the lens of fiscal redistribution and public investment. Property tax, being progressive and recurrent, enables the government to collect more revenue from higher-value assets and channel these funds into public goods such as infrastructure, education, and healthcare, all of which directly enhance productivity, social welfare, and overall living standards. In this way, a well-designed property tax system supports inclusive economic growth, ensuring that wealth generated in the property market contributes to wider societal development.

Between 2010 and 2014, rising property taxes coincided with sustained GDP growth, reflecting the government's ability to reinvest fiscal gains from the property sector into productive areas of the economy. The brief stagnation in tax growth between 2015 and 2019 was accompanied by slower gains in GDP per capita, suggesting that restrained property tax inflows may have limited fiscal space for public spending. From 2020 onward, both property tax and GDP per capita surged sharply, driven by higher property valuations and progressive tax reforms, alongside a

strong post-pandemic economic rebound. This simultaneous rise indicates that increased property taxation can reinforce macroeconomic resilience, providing the government with stable, countercyclical revenue to fund recovery efforts and social programs.

4.6 Relationship Between Landed Property Tax and HDI

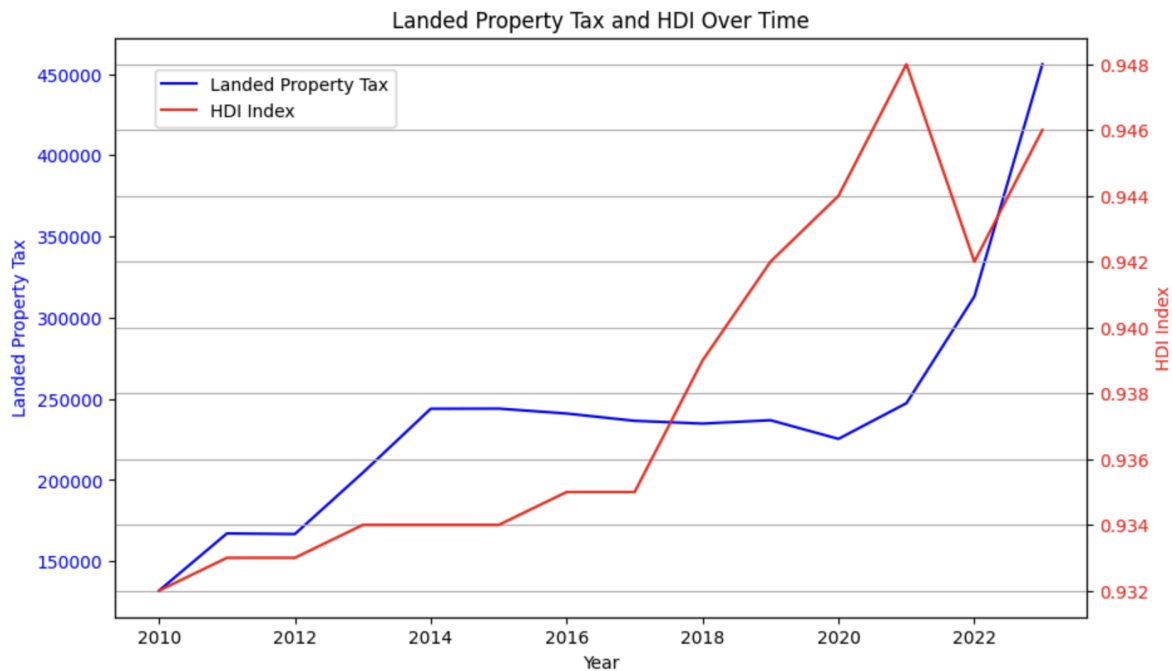


Figure 6: Correlation Between Property Tax and HDI Over Time

Figure 6 indicates a positive relationship between landed property tax revenue and Singapore’s Human Development Index (HDI) from 2010 to 2023. Both indicators rise steadily over time, suggesting that increases in property tax coincide with improvements in the nation’s standard of living. Similar to the above explanation, this can be explained by the fiscal redistribution effect of property taxation. Higher property taxes, particularly from the landed segment, provide the government with additional resources to invest in public infrastructure, education, healthcare, and social programs, the very components used to calculate HDI. By capturing a portion of wealth from high-value assets and reallocating it toward social development, property tax contributes indirectly to greater equity and higher overall human welfare.

4.7 Instrumental Variable Method as Causation Determinant

To ensure the correct direction of causation between property tax and housing affordability, an instrumental variable (IV) approach is adopted. Government debt is used as the instrument for property tax because it is plausibly correlated with fiscal pressures that influence tax decisions, higher public debt often leads governments to increase or adjust tax revenues, thereby satisfying the relevance condition. At the same time, government debt is not directly linked to housing affordability in the short run, as it does not immediately alter housing supply, demand, or household-level purchasing power, thus meeting the exclusion restriction. By leveraging government debt as an IV, we isolate the exogenous variation in property tax, allowing for a cleaner and more credible estimate of its causal effect on housing affordability.

The 2 linear equations that we are going to use are:

$$\text{PropertyTax}_t = \alpha + \beta_1 \text{GovDebt}_t + \beta_2 \text{Year}_t + u_t$$

$$\text{Affordability}_t = \gamma + \delta_1 \widehat{\text{PropertyTax}}_t + \delta_2 \text{Year}_t + \epsilon_t$$

2SLS Regression Results for Landed Properties:

IV-2SLS Estimation Summary

Dep. Variable:	price_income_ratio	R-squared:	-1.2690
Estimator:	IV-2SLS	Adj. R-squared:	-1.6815
No. Observations:	14	F-statistic:	6.3670
Date:	Mon, Nov 10 2025	P-value (F-stat)	0.0414
Time:	03:12:12	Distribution:	chi2(2)
Cov. Estimator:	robust		

Parameter Estimates

	Parameter	Std. Err.	T-stat	P-value	Lower CI	Upper CI
const	527.51	232.79	2.2660	0.0234	71.252	983.76
year	-0.2612	0.1165	-2.2425	0.0249	-0.4894	-0.0329
property_tax	1.538e-05	9.321e-06	1.6504	0.0989	-2.885e-06	3.365e-05

Figure 7: Regression Results for Landed Properties

This positive coefficient indicates that an increase in property tax is associated with a very slight increase in the price-to-income ratio, implying lower affordability. However, the relationship is not statistically strong (significant only at the 10% level).

4.8 Relationship between Lagged GDP per capita and property tax

Pearson correlation coefficient between Lagged GDP per capita and Landed Property Tax: 0.8794781751142876
Pearson correlation coefficient between Lagged GDP per capita and Condo Property Tax: 0.8810005027491662

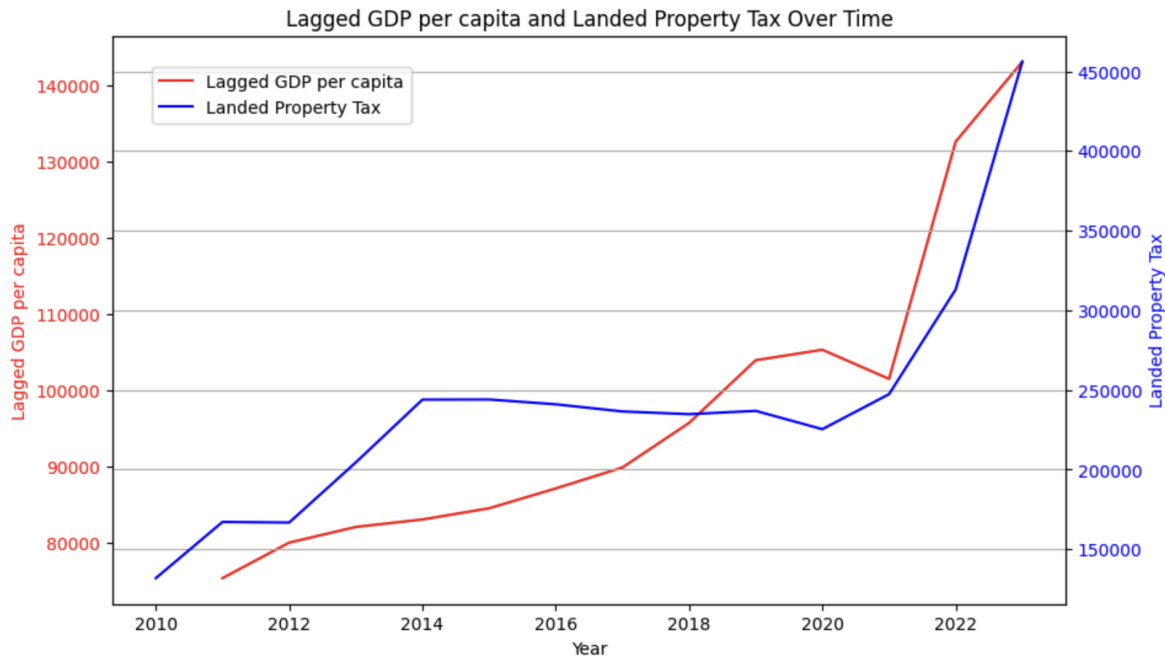


Figure 8: Lagged GDP per capita and Property Tax Revenue Over Time

Using lagged variables provides clearer evidence of the relationship between economic growth and property taxation. The strong positive correlations (0.879 for landed properties and 0.881 for condominiums) indicate that increases in GDP per capita are consistently followed by higher property tax revenues. This suggests that rising national income levels tend to drive subsequent growth in property values and tax collections.

4.9 Relationship between Lagged GDP per Capita and HDI

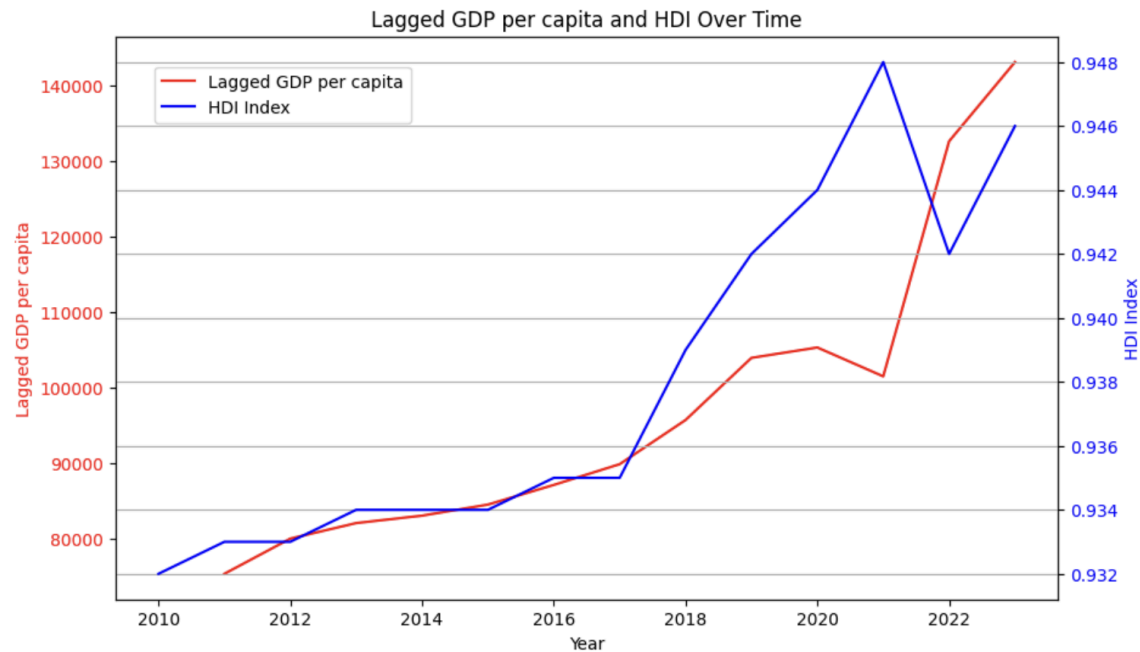


Figure 9: Lagged GDP per capita and HDI Over Time

The relationship between lagged GDP per capita and Singapore's Human Development Index (HDI) shows a clear, positive, and mutually reinforcing trend over time. As lagged GDP per capita increases, the HDI consistently improves, reflecting how broader economic growth contributes to higher living standards, better education, and improved health outcomes. The parallel upward movement of both variables suggests that rising national income in earlier periods is closely associated with subsequent improvements in overall human development.

5. Overall Insights and Final Conclusion

Overall, the findings reveal that property taxation in Singapore functions more as a fiscal and redistributive tool than as a direct lever for housing affordability. Across both landed and condominium segments, tax increases show only weak causal effects on the price-to-income ratio, suggesting that affordability is largely shaped by deeper structural factors such as land scarcity, household income growth, and wealth-driven demand. At the same time, the positive co-movement between property tax, GDP per capita, and HDI underscores the broader role of taxation in supporting public investment and social welfare, linking fiscal capacity to improvements in overall living standards rather than to immediate price moderation (*IRAS | the Singapore Tax System*, n.d.).

Economically, this implies that property tax is more effective at sustaining long-term equity and fiscal stability than at correcting short-run affordability pressures. Macroprudential tools like the TDSR and ABSD remain more impactful in curbing speculative demand, while progressive tax reforms ensure that wealthier households contribute proportionally more to national development. To improve housing accessibility, tax policy should thus work in tandem with supply-side measures, such as land release, redevelopment incentives, and build-to-rent programs, to address the root causes of price escalation.

6. Possible Limitations

Furthermore, several limitations in our methodology have been identified, which may have hindered the levels of validity as well as accuracy of the whole study. First of all, the utilisation of property tax revenue instead of its rates during this study has resulted in the data being a combined reflection of both policy actions and economic performance. Therefore, it became difficult to distinguish whether the tax revenue fluctuations are caused by deliberate fiscal adjustments or by general economic conditions, such as income growth or higher demands for housing. This difficulty is also applied to the different indicators that have been compared to the property tax revenue. Hence, proper causality could not be established between property tax revenues and the variables examined.

Secondly, the price-to-income ratio method used to measure housing affordability was also not the best option. Solely using income as an indicator of one's purchasing power goes against the goal of making this study a holistic approach in picturing the housing market in Singapore, as there are multiple determinants that could well impact the consumers' purchasing power within the market, such as mortgage, grant subsidies, and loans. Additionally, with the price of other daily products like food items increasing due to rising inflations in Singapore (Lee et al., 2024), the ratio might not essentially capture the broad view of the financial pressure, and therefore the housing affordability, that Singaporeans are currently facing despite the stable income growth observed.

Hence, while the results found in this study may have revealed meaningful associations between property tax revenue and other economic indicators from affordability to well-being, it is crucial that one keeps a mindful stance when noting down insights from this study. This consideration stems from the fact that the results do not fully capture the complex, multidimensional relationships between the observed variables in the market.

7. Future Considerations

In response to the limitations, one of the future considerations that shall be taken into account for future research would be to employ broader, yet more targeted indicators of the economy. For instance, instead of fixing the focus only on price-to-income ratio and GDP per capita, variables such as interest rates, household debt levels, and housing satisfaction measures could be incorporated. Incorporating such targeted indicators would be able to offer a more comprehensive understanding of how government fiscal tools could affect the macroeconomic outcomes and the people's quality of everyday life.

Moreover, if given more time and resources, on top of expanding the dataset, future research could apply richer econometric techniques like panel data, structural VAR models, or difference-in-differences to capture dynamic policy effects. As mentioned above, incorporating more granular data on income distribution, property ownership type, and housing completions using more robust econometric methods would allow a deeper understanding of how taxes, credit policies, and supply constraints interact. Exploring behavioural and spatial dimensions,

such as buyer expectations, foreign investment flows, and regional heterogeneity, would also strengthen the causal story and inform more targeted, evidence-based housing policies.

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